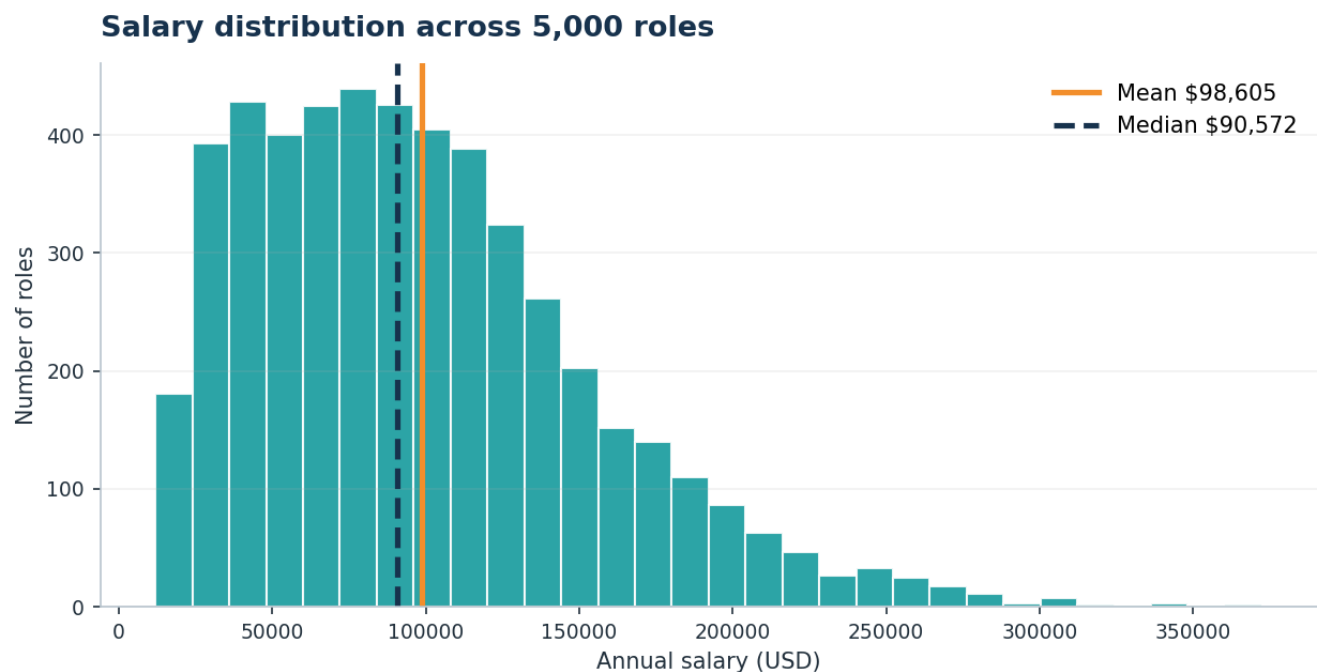


AI & Data Science Job Market Analysis

A 5,000-record exploration of compensation, career structure, skills, work models and AI adoption

\$98,605	5,000	\$90,572	70.3%
Average salary	Records	Median salary	Daily AI use

This report analyzes 5,000 observations across 27 variables. The dataset covers job titles, experience, employment structure, company characteristics, location, education, programming language, AI adoption, compensation and employee sentiment.



Key takeaways

- Average salary: \$98,605; median salary: \$90,572.
- Salary spans \$12,000 to \$372,347.
- The analysis is descriptive: relationships are not proof of causation.

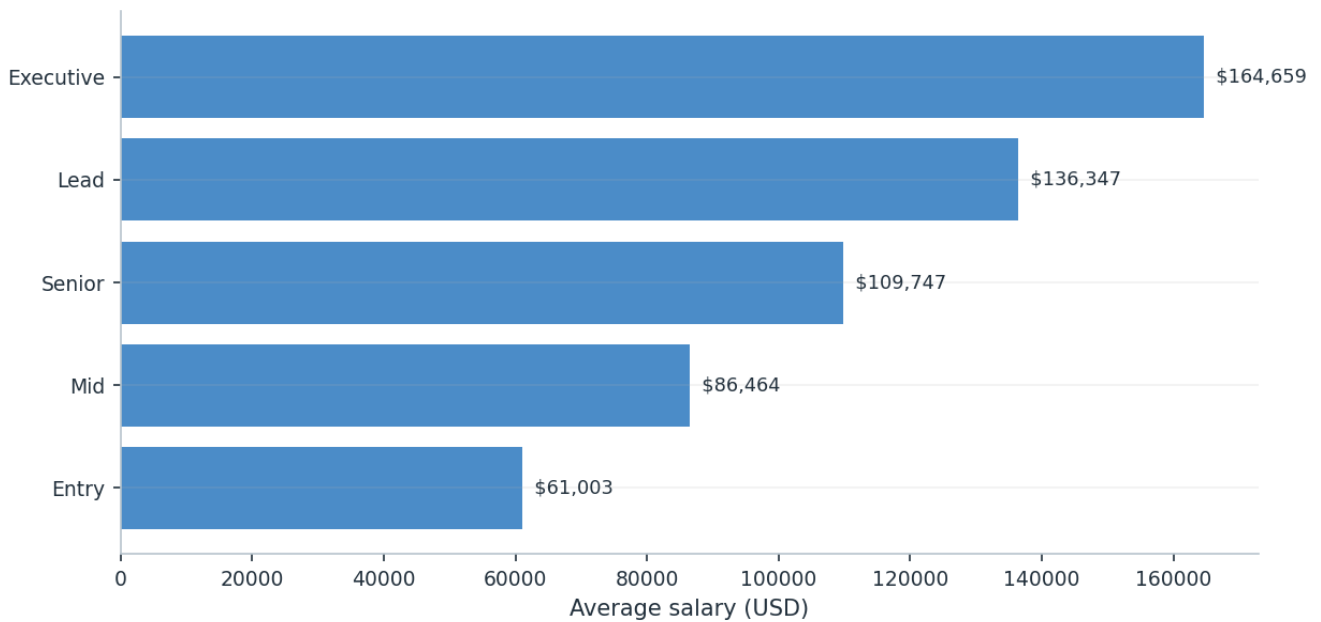
Source: supplied ai_ds_job_salaries_2026.csv and the accompanying analysis notebook. All monetary figures are in USD. Prepared as a shareable analytical report.

Executive Summary

The strongest story is that seniority, role specialization, location and management responsibility matter most

Compensation increases consistently across the experience ladder, from about \$61,003 for entry-level roles to \$164,659 for executive roles. Beyond experience, AI/ML-focused titles, people management and company location are prominent differentiators.

Average salary rises sharply with experience



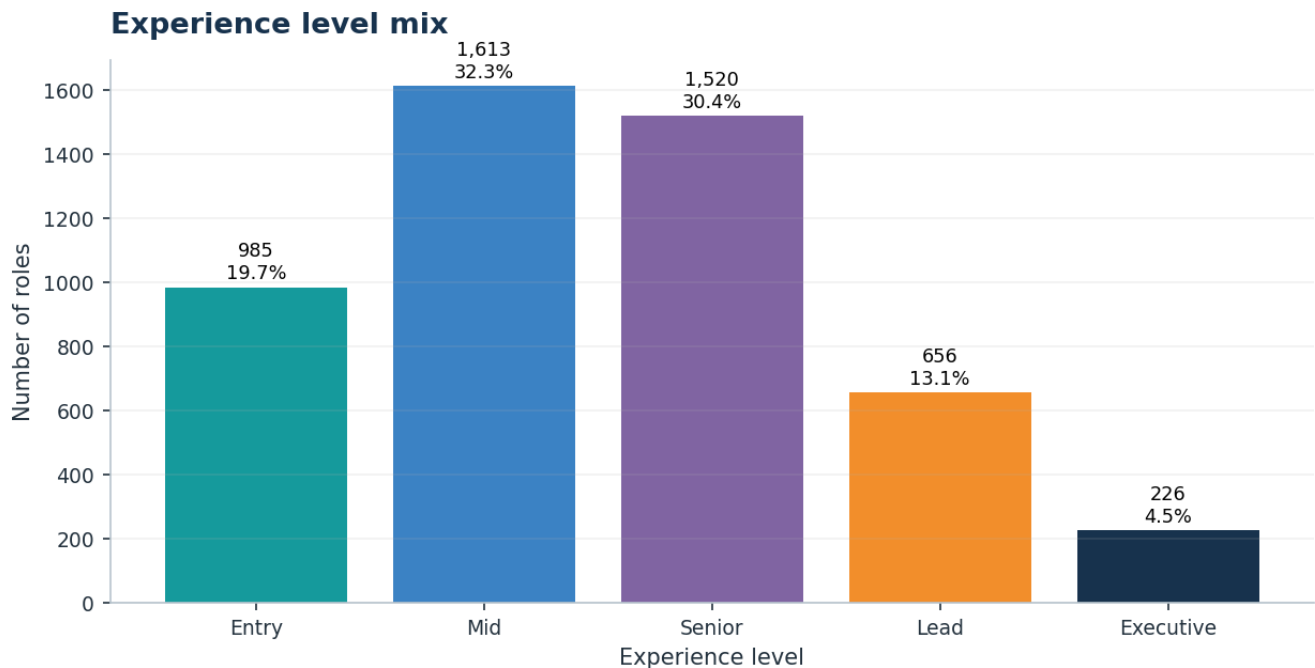
Key takeaways

- Experience is the strongest numeric predictor in the dataset (correlation ≈ 0.47).
- Roles with ML in the title average \$110,246, versus \$91,897 without ML.
- People managers average \$130,987, versus \$89,042 for non-managers.

Dataset Profile & Analytical Approach

What is being measured and how the analysis should be interpreted

The dataset contains 5,000 complete records and 27 columns spanning categorical, boolean and numeric variables. The notebook supplied with the data focuses heavily on distribution charts and group comparisons; this report extends that approach with salary relationships, correlations and practical interpretation.



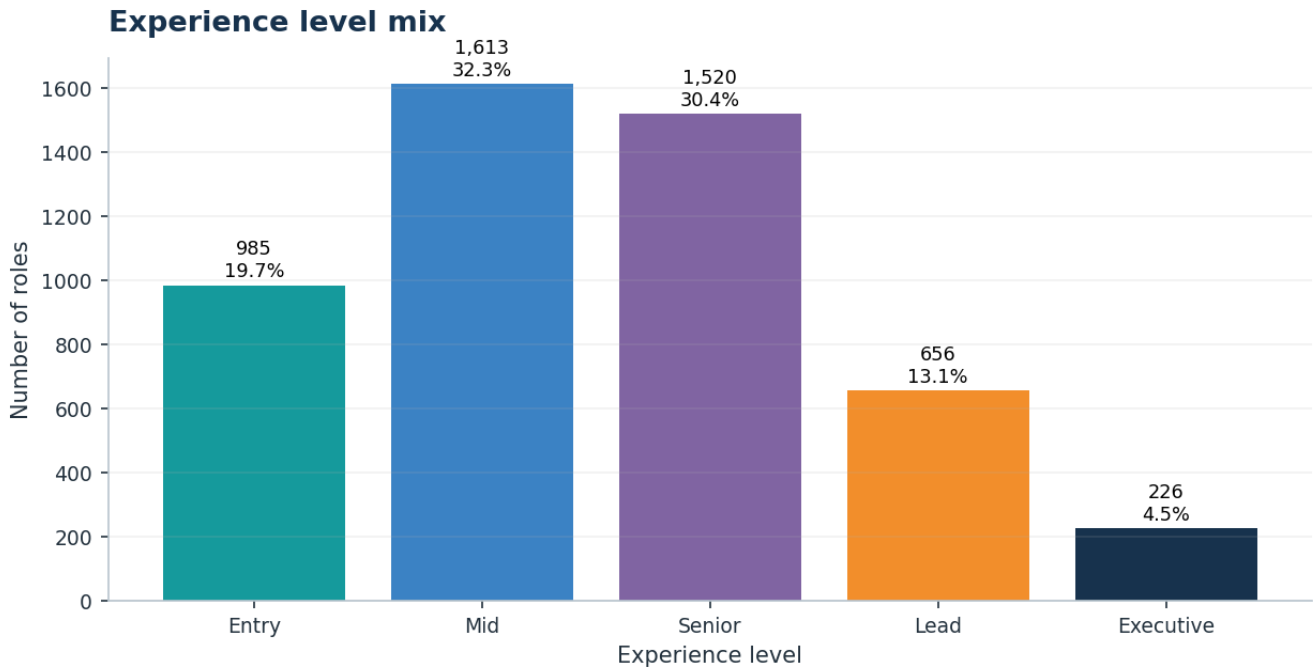
Key takeaways

- No missing values were found in the supplied CSV.
- Salary is already expressed in USD, making cross-location comparisons straightforward within this dataset.
- Grouped averages show association; they should not be interpreted as causal effects.

Overall Experience Mix

Mid and senior roles make up the majority of observations

The sample is weighted toward mid and senior professionals, with fewer executive observations. This matters because overall salary averages reflect the composition of the sample as well as underlying pay differences.



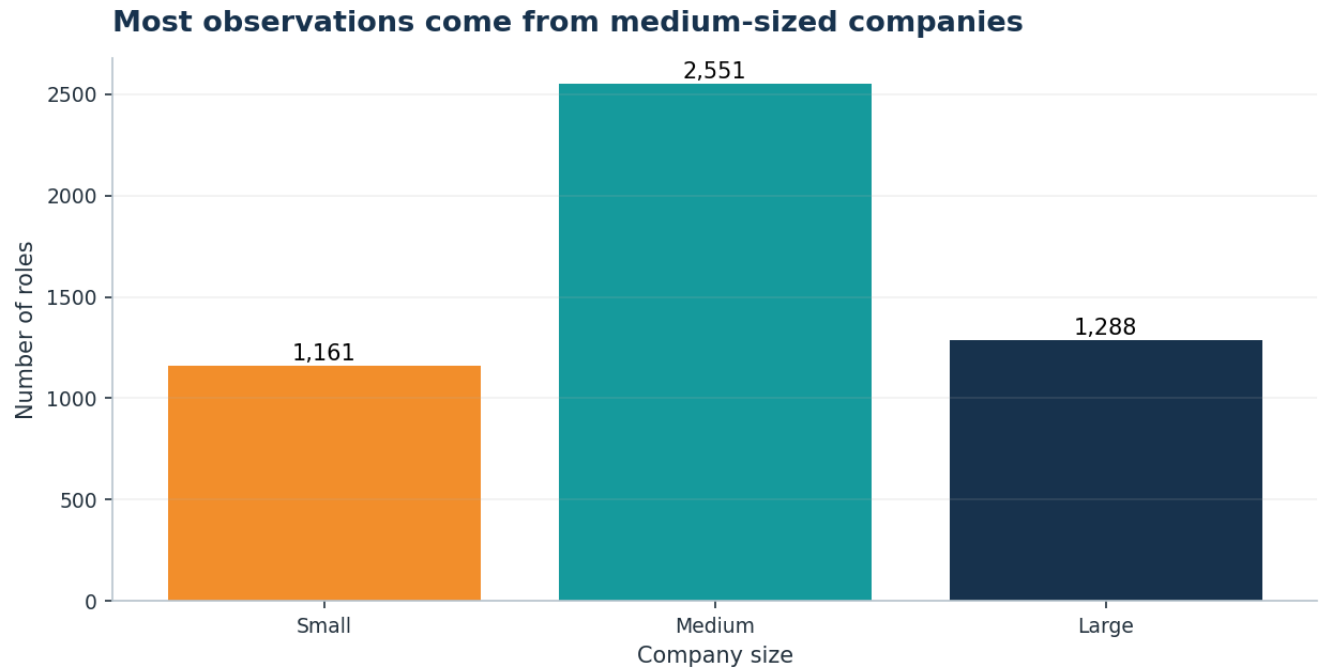
Key takeaways

- Mid-level roles: 1,613 observations.
- Senior roles: 1,520 observations.
- Executive roles are relatively scarce, so their average is more sensitive to individual observations.

Company Size

Medium-sized companies are the largest group in the dataset

Medium companies account for just over half of the records, while small and large organizations form smaller but substantial groups. Company size is therefore useful for comparing pay and work conditions without letting one size dominate the sample completely.



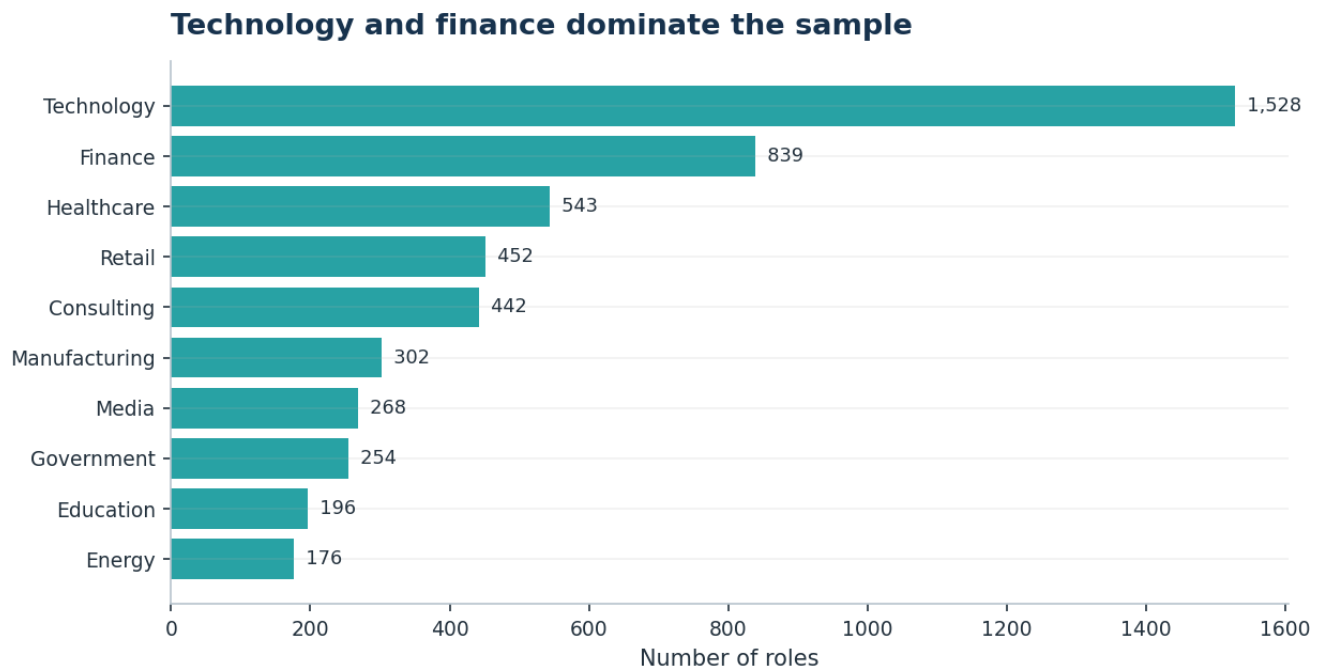
Key takeaways

- Medium companies: 2,551 roles.
- Large companies have the highest average salary at \$113,401.
- Small-company pay is lower on average, but company size alone does not explain the full salary range.

Industry Composition

Technology and finance are prominent sectors

Technology is the largest industry group, followed by finance and healthcare. Smaller sectors such as education and energy have fewer observations, so their estimates should be read with more caution.



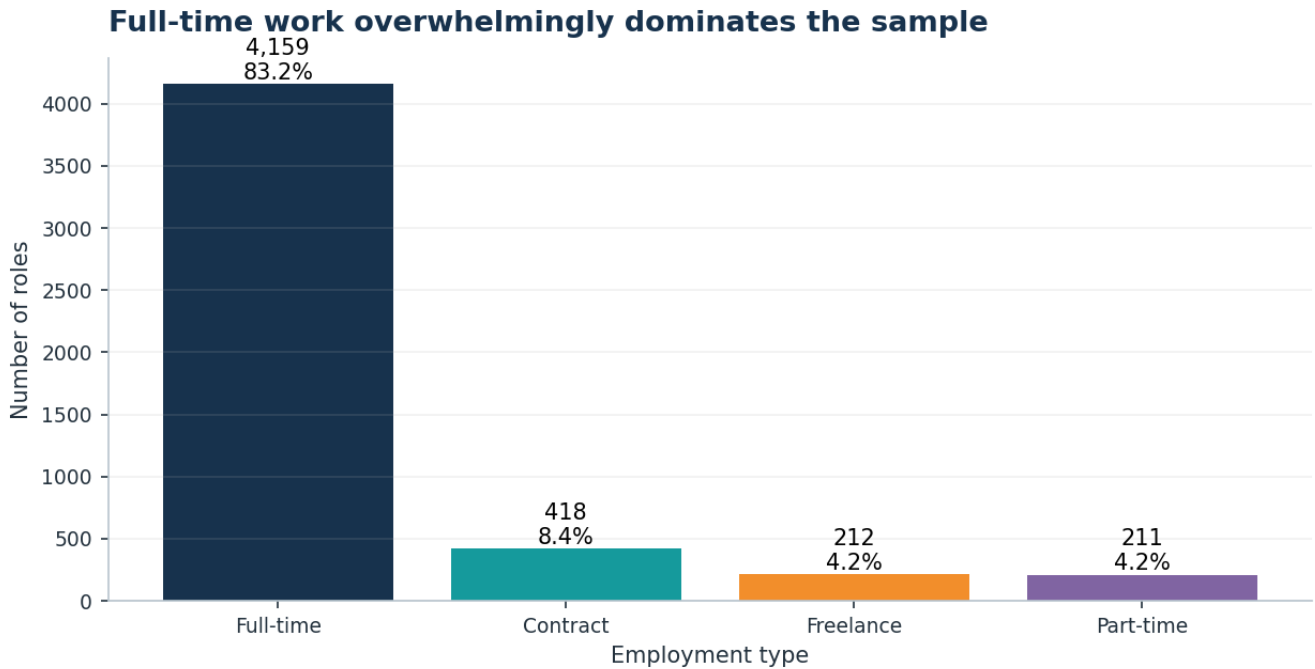
Key takeaways

- Technology contributes 1,528 records.
- Finance has the highest industry average salary at \$111,842.
- Industry composition affects the overall salary benchmark because sectors have different role mixes and pay structures.

Employment Type Mix

The dataset is overwhelmingly full-time

Full-time employment represents the vast majority of the observations. Contract, freelance and part-time roles form smaller segments, which limits how confidently their averages can be generalized.



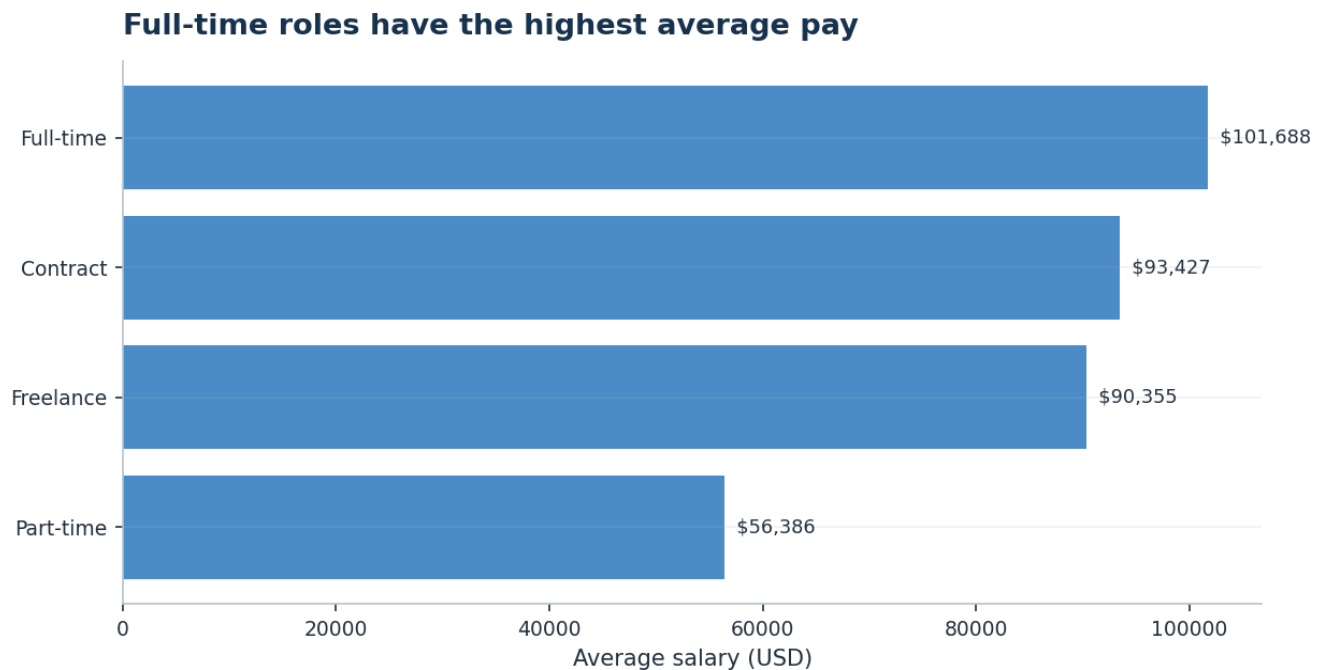
Key takeaways

- Full-time roles: 4,159 (83.2%).
- Part-time work has the lowest observed average salary.
- The dominance of full-time roles means the headline salary benchmark mainly reflects permanent employment.

Salary by Employment Type

Employment structure is associated with meaningful pay differences

Full-time roles have the highest average compensation in this sample, while part-time roles are substantially lower. Contract and freelance averages sit between these extremes.



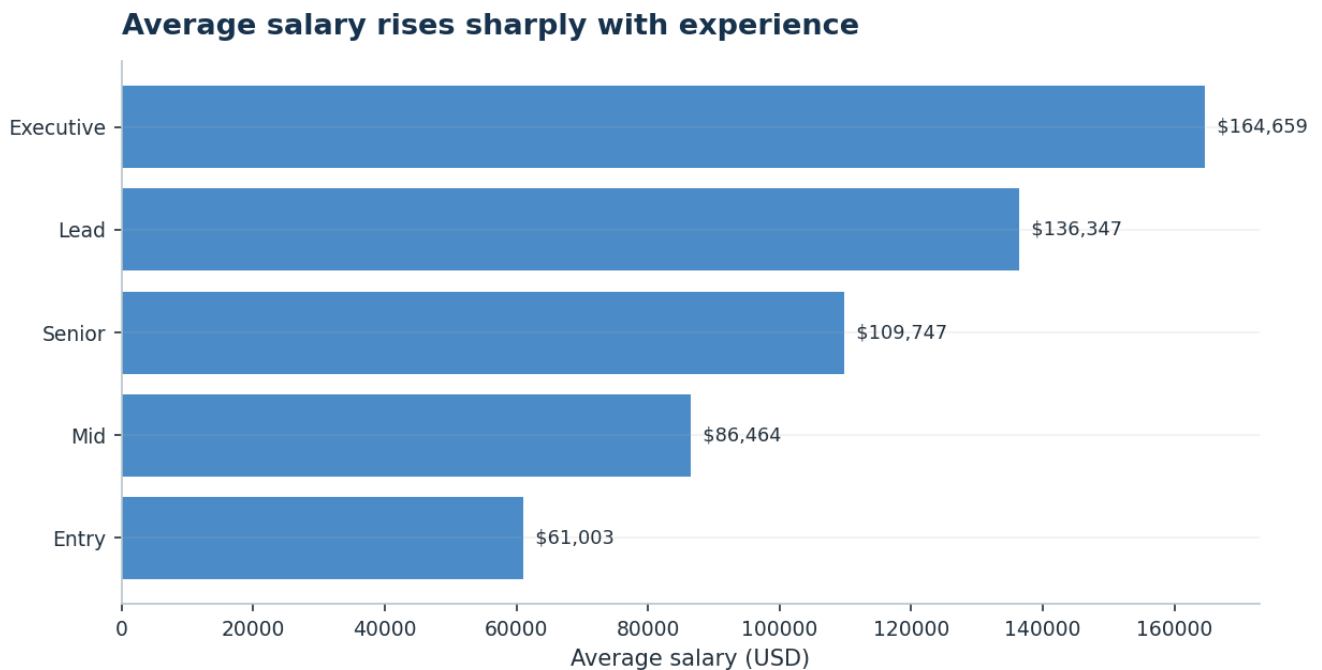
Key takeaways

- Full-time average: \$101,688.
- Part-time average: \$56,386.
- Employment type should be considered alongside hours, seniority and role when making compensation comparisons.

Salary by Experience Level

Career progression is one of the clearest salary gradients

Average salary rises at every step from entry through executive level. The pattern is monotonic, making experience level one of the most useful segmentation variables in the dataset.



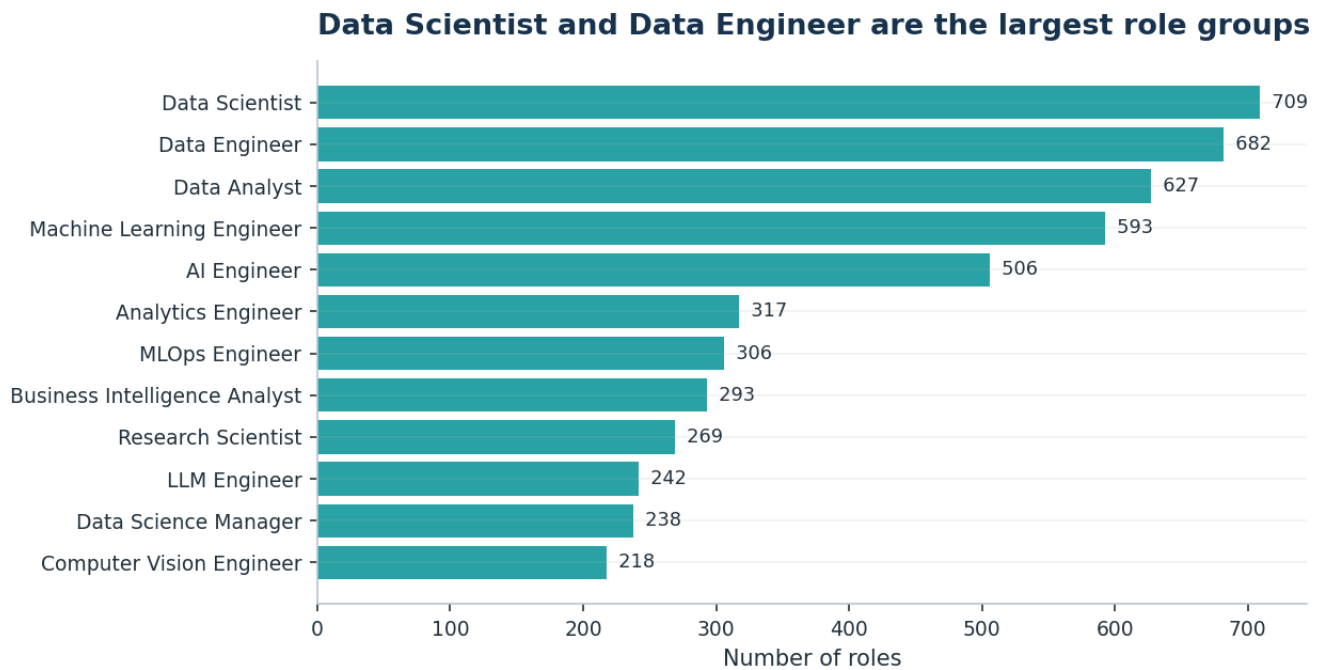
Key takeaways

- Entry average: \$61,003.
- Senior average: \$109,747.
- Executive average: \$164,659.
- The executive-to-entry average salary is roughly 2.7×.

Job Title Composition

The sample contains a broad mix of analytics, engineering, science and management roles

Data Scientist, Data Engineer and Data Analyst are the largest individual job-title groups. This creates a useful cross-section of the modern data/AI workforce rather than focusing on a single occupation.



Key takeaways

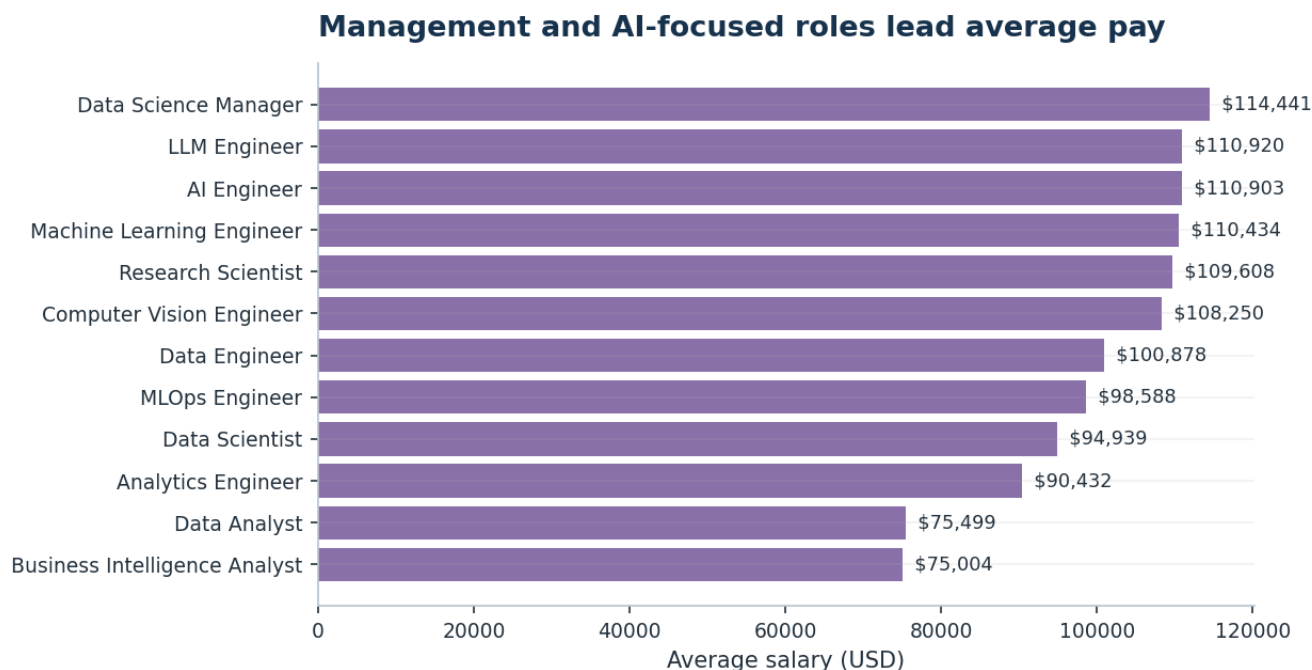
- Data Scientist is the largest group with 709 records.
- Data Engineer follows with 682.
- The dataset contains both technical specialist roles and management-oriented roles.

Salary by Job Title

Specialized AI/ML and management roles sit near the top

Data Science Manager has the highest average salary among the listed job titles at about \$114,441.

Analyst-oriented titles are lower, while engineering, research and management positions generally sit higher.



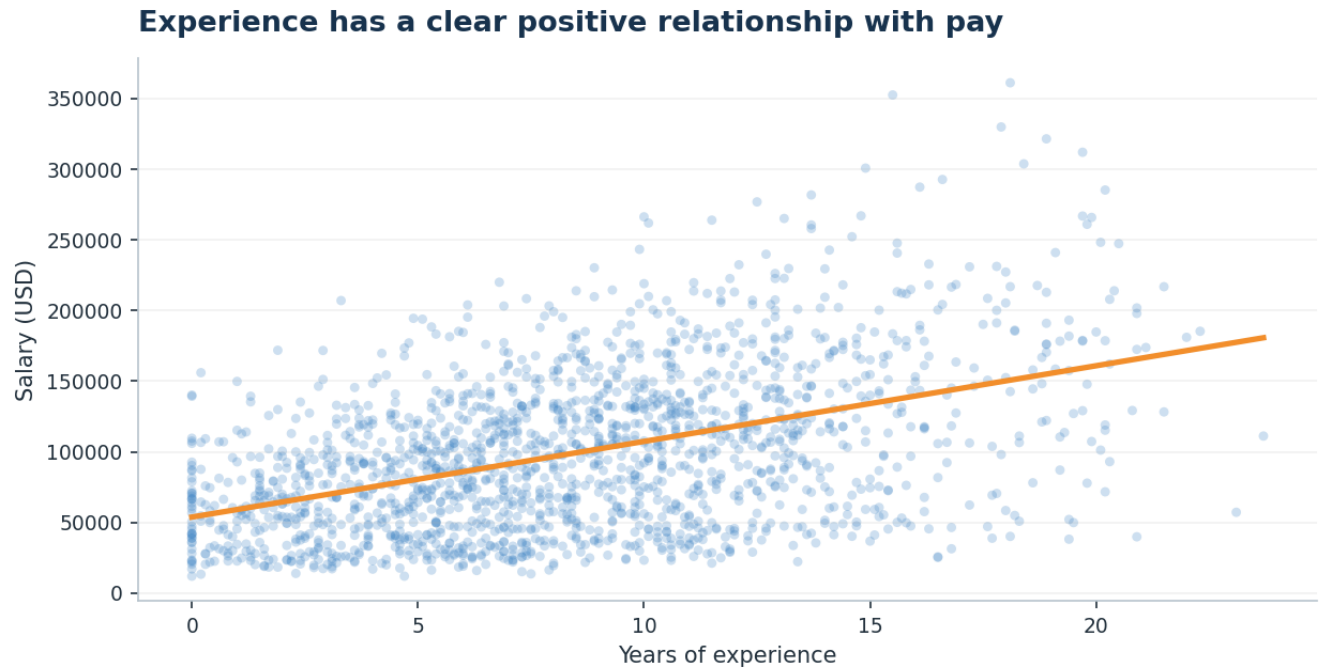
Key takeaways

- Highest average: Data Science Manager at \$114,441.
- Data Analyst average: \$75,499.
- Business Intelligence Analyst average: \$75,004.
- Role specialization appears to matter substantially, although role mix and experience also contribute.

Years of Experience vs Salary

The relationship is positive, but experience does not determine salary perfectly

The scatter plot shows a clear upward trend between years of experience and salary. The wide vertical spread at many experience levels indicates that location, role, industry and other factors also create substantial differences.



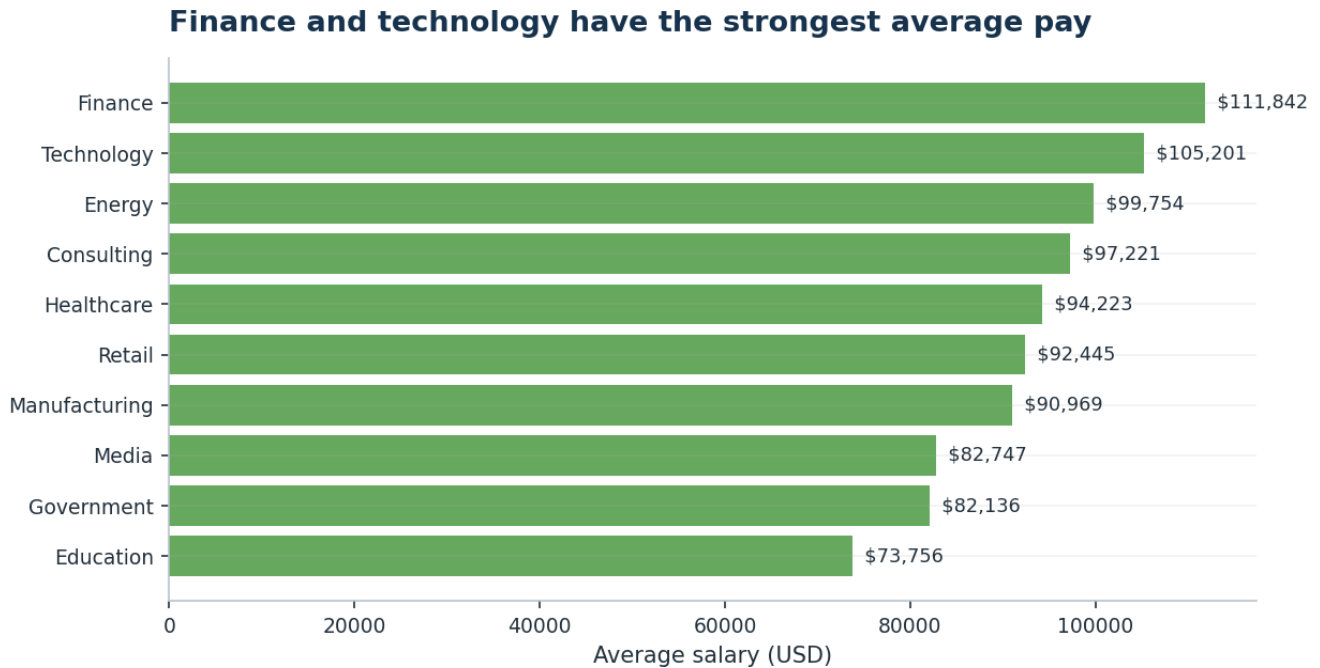
Key takeaways

- Pearson correlation between years of experience and salary: 0.47.
- Experience explains directionally more salary variation than AI-tool usage or certifications.
- The relationship should not be interpreted as a fixed salary increase per year.

Salary by Industry

Finance and technology lead the observed industry averages

Finance records the highest average salary, followed by technology. Education and government are among the lower-paying sectors in this sample. These differences may reflect role mix, geography and sector-specific compensation structures.



Key takeaways

- Finance average: \$111,842.
- Technology average: \$105,201.
- Education average: \$73,756.
- Industry is a useful market-level segmentation variable.

Salary by Company Location

Geography is one of the largest observed salary differentiators

US-based company locations have the highest average salary in the dataset at roughly \$135,662. India and Brazil are much lower on a USD basis. This is a descriptive comparison of the supplied data, not a cost-of-living adjusted analysis.

Company location is associated with a very large pay gap



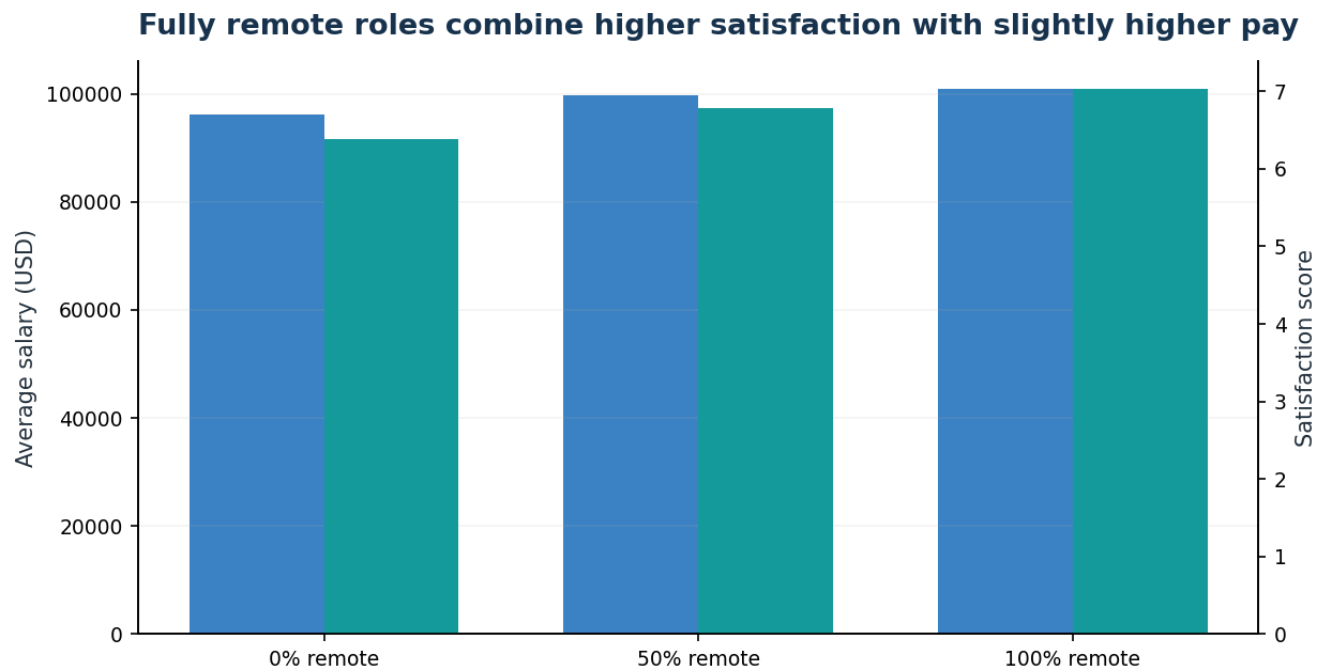
Key takeaways

- Highest company-location average: US at \$135,662.
- US average: \$135,662.
- India average: \$37,577.
- Salary comparisons across countries should ideally be paired with purchasing-power or cost-of-living context.

Remote Work and Compensation

Fully remote roles show the highest satisfaction and slightly higher average pay

The dataset separates fully on-site, hybrid and fully remote roles. Fully remote workers have the highest average satisfaction score and a modestly higher salary average than hybrid or on-site workers.



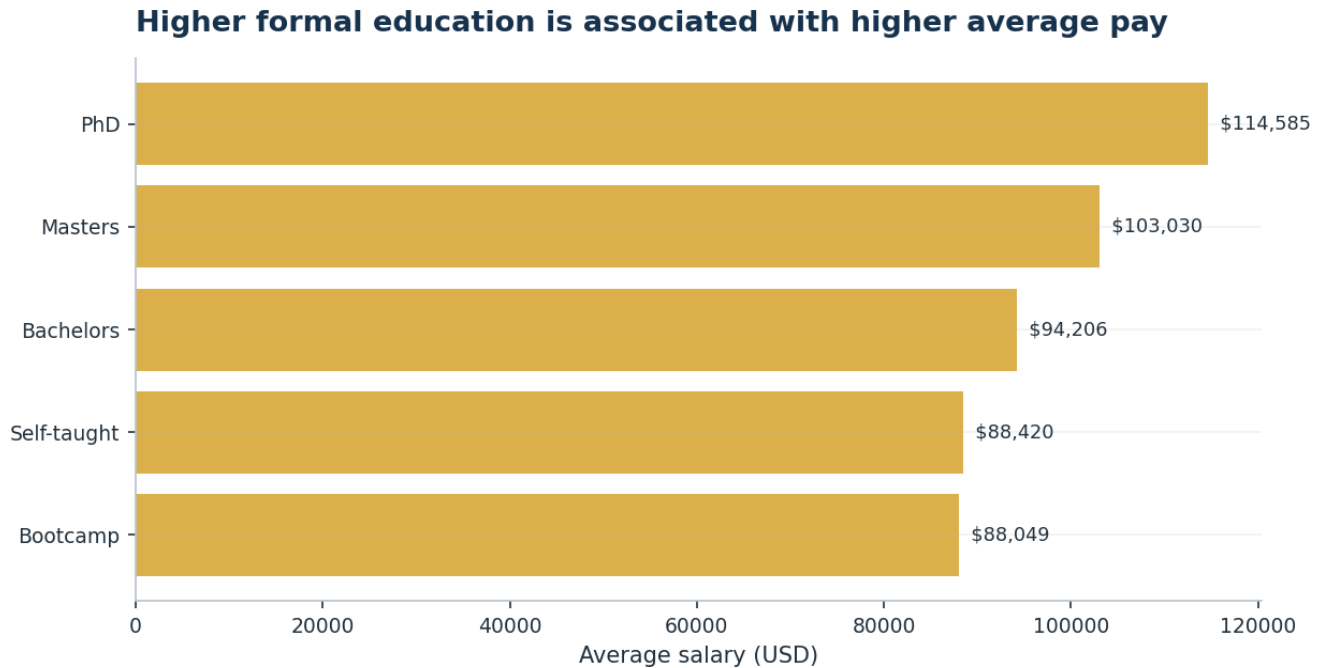
Key takeaways

- Fully remote average salary: \$100,996.
- Fully remote satisfaction: 7.04/10.
- On-site satisfaction: 6.38/10.
- The difference is an association and may partly reflect role, seniority or employer composition.

Education and Salary

Higher academic credentials are associated with higher average compensation

PhD holders have the highest average salary, followed by master's and bachelor's graduates. Self-taught and bootcamp groups have lower averages in this sample, but these groups also differ in role and experience composition.



Key takeaways

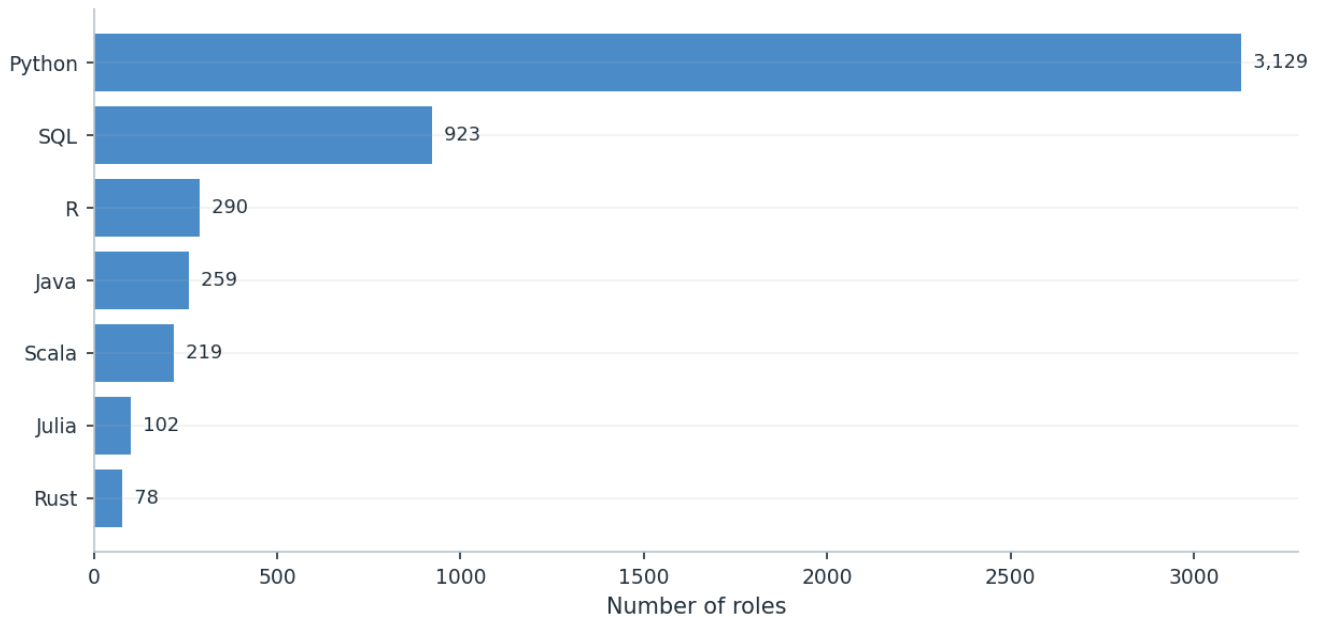
- PhD average: \$114,585.
- Masters average: \$103,030.
- Bachelors average: \$94,206.
- Education should be treated as one signal among several rather than a guarantee of compensation.

Primary Programming Language

Python dominates the sample, while average salaries are comparatively close across major languages

Python is by far the most common primary language, followed by SQL and Java. Frequency is much more uneven than salary: the most common languages do not automatically have the highest average salary.

Python is the dominant primary language in the dataset



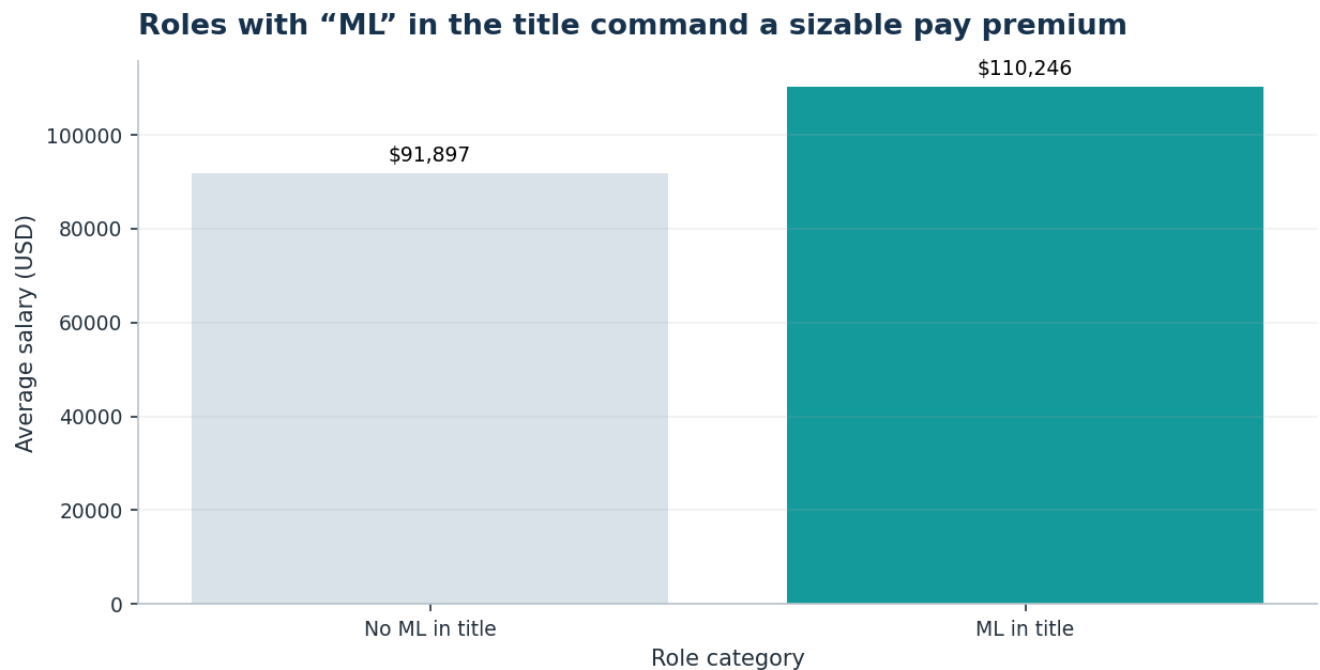
Key takeaways

- Python appears in 3,129 roles.
- Scala has the highest language-level average salary at \$102,419.
- Language choice is likely confounded by job title, experience and industry.

ML in the Job Title

ML-branded roles command a substantial salary premium in this dataset

Roles whose titles explicitly contain an ML designation have noticeably higher average compensation than roles without ML in the title. This likely captures specialization and role seniority together.



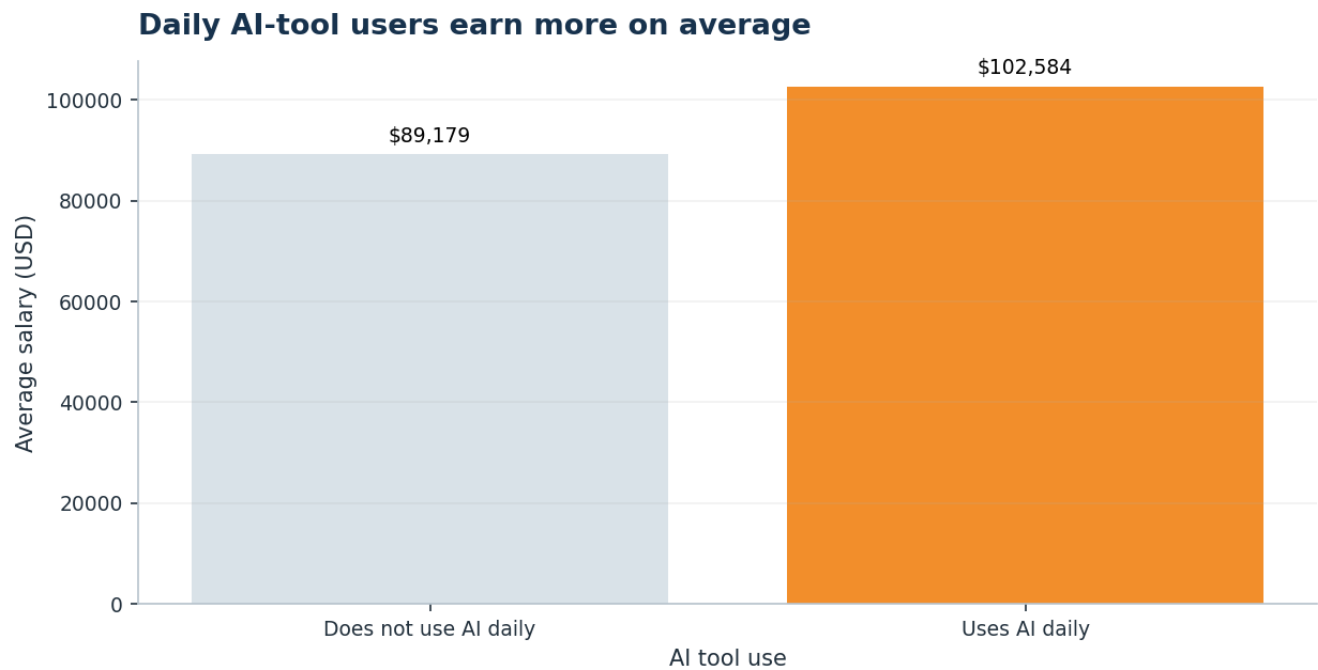
Key takeaways

- ML-title roles average \$110,246.
- Non-ML-title roles average \$91,897.
- The observed gap is about \$18,350.
- The title itself is a label; it does not prove that ML caused the salary difference.

Daily AI Tool Adoption

Most professionals in the sample use AI tools daily

About seven in ten observations report daily AI-tool use. Daily users have a higher average salary, while their average satisfaction is very similar to non-users.



Key takeaways

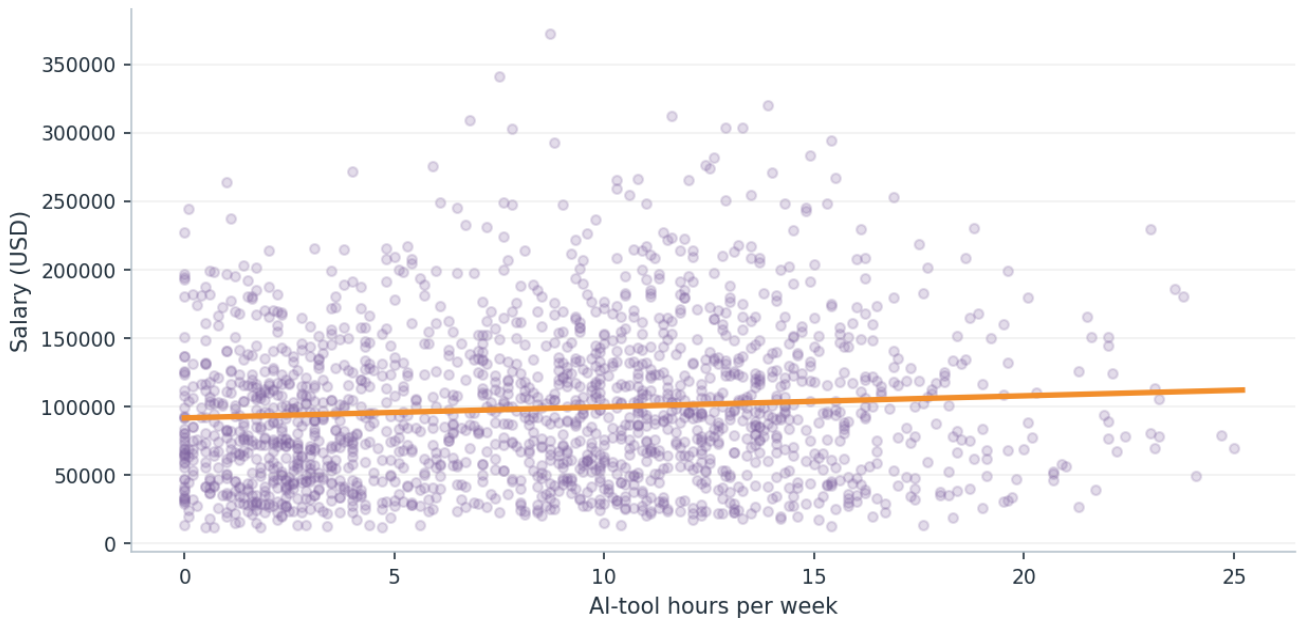
- Daily AI-tool adoption: 70.3%.
- Daily users average \$102,584.
- Non-daily users average \$89,179.
- AI adoption appears widespread rather than limited to a small specialist group.

AI Usage Intensity vs Salary

Using AI more hours per week is associated with only a weak salary relationship

The relationship between AI-tool hours per week and salary is positive but weak. This is important: simply spending more time with AI tools is not nearly as strong a salary signal as experience or role specialization.

AI usage intensity has a weaker relationship with salary than experience



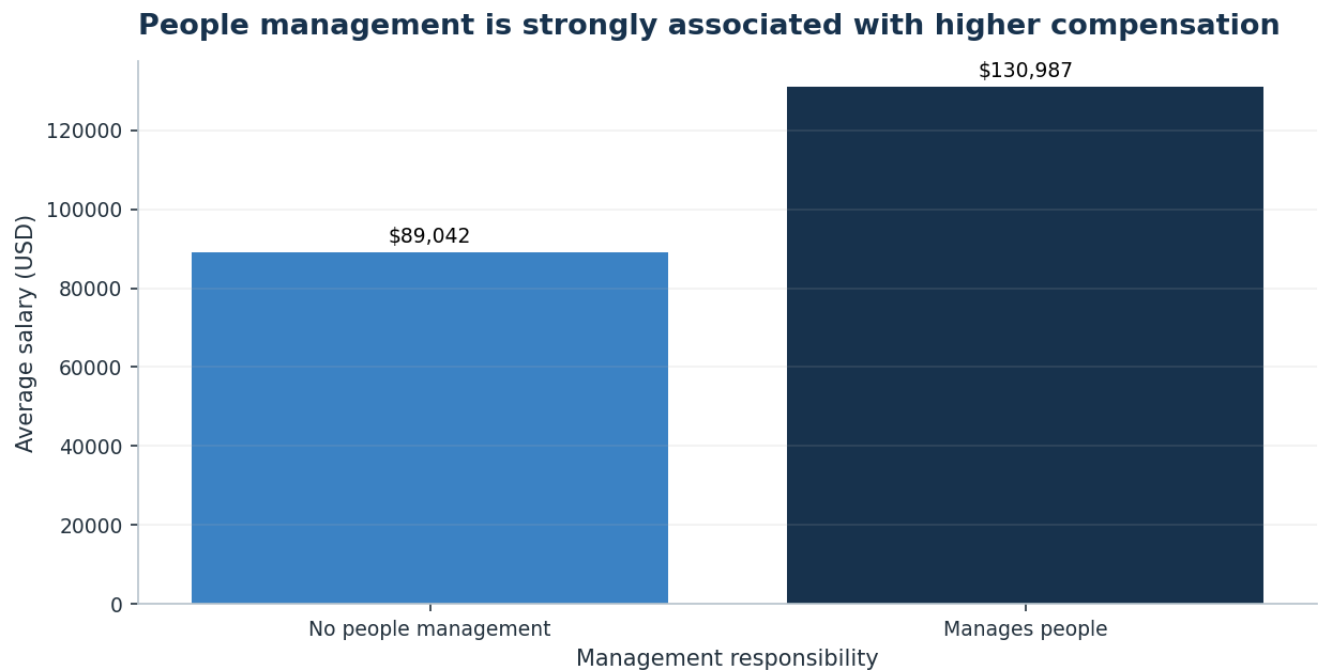
Key takeaways

- Correlation of AI-tool hours with salary: 0.08.
- Tool adoption and tool intensity are different concepts.
- Higher-paid roles may have greater AI exposure without the tools themselves being the direct cause of higher pay.

People Management and Pay

Management responsibility is strongly associated with compensation

The 22.8% of records that manage people have substantially higher average salaries. Management responsibility is likely a proxy for scope, seniority and organizational accountability.



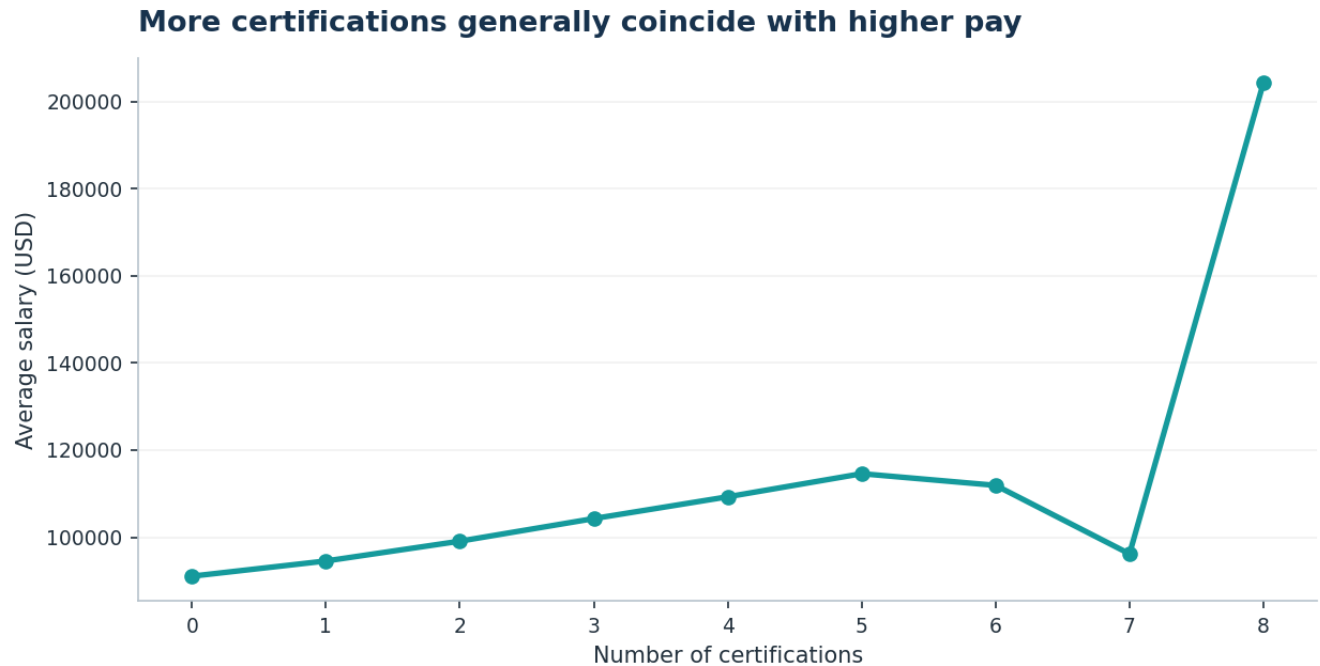
Key takeaways

- Managers average \$130,987.
- Non-managers average \$89,042.
- The average gap is about \$41,945.
- Leadership scope appears to be financially rewarded in the observed sample.

Certifications and Salary

Certification count generally moves with higher average salary, with diminishing stability at the extremes

Average salary generally increases from zero through several certifications. Very high certification counts are rare, so the final points should not be treated as stable population estimates.



Key takeaways

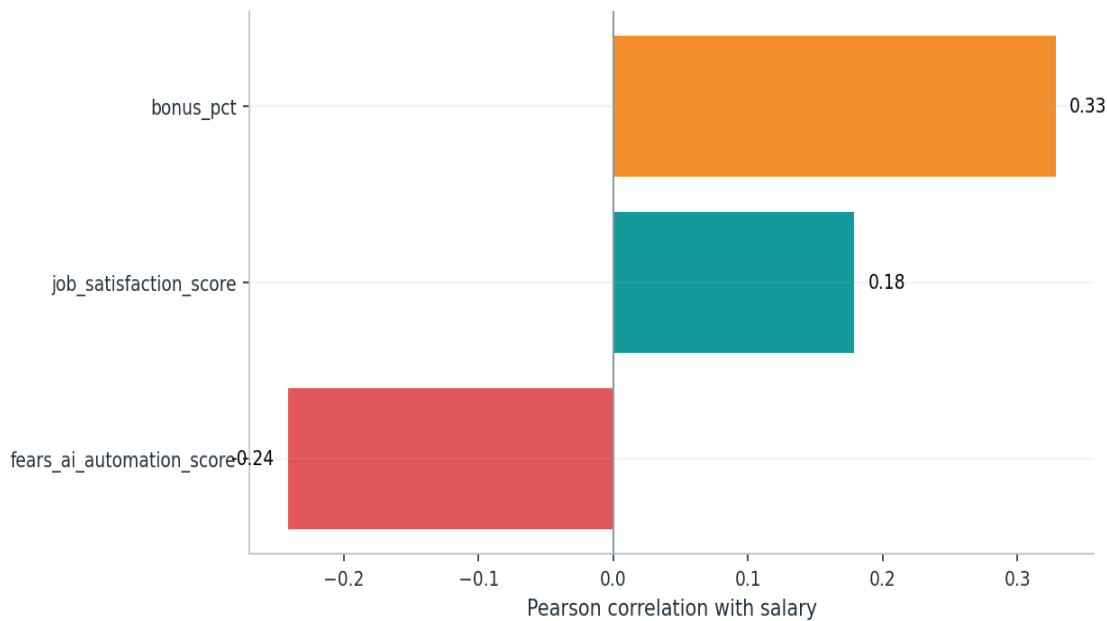
- The broad pattern is positive from 0 to 5 certifications.
- 0 certifications average \$91,066.
- 5 certifications average \$114,587.
- Certifications may signal commitment or seniority rather than directly causing salary growth.

Salary, Bonus, Satisfaction and AI-Automation Fear

Financial variables and sentiment show different strengths of association

Salary has the strongest positive correlation among these variables with bonus percentage, followed by job satisfaction. Fear of AI automation has a moderate negative correlation with salary, suggesting that higher-paid roles in this sample tend to report lower automation concern.

Salary correlates most with bonus, satisfaction and lower AI-automation fear



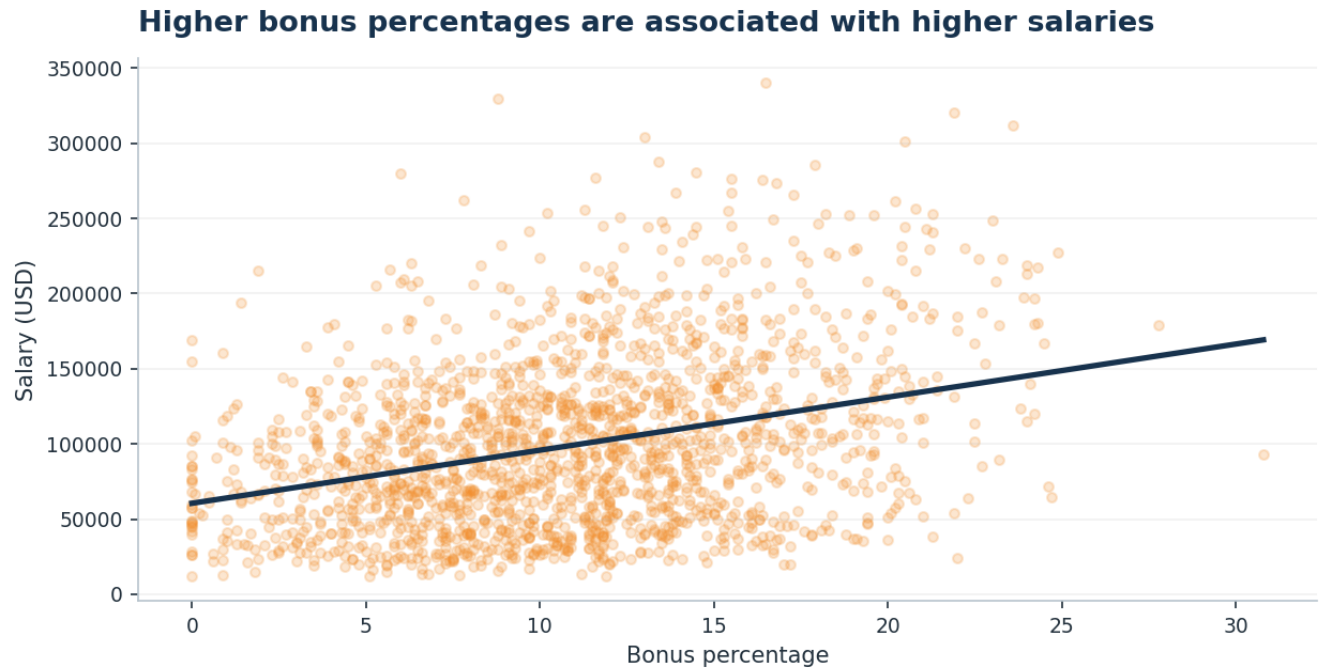
Key takeaways

- Bonus-to-salary correlation: 0.33.
- Satisfaction-to-salary correlation: 0.18.
- Automation-fear-to-salary correlation: -0.24.
- Correlation measures association, not causation.

Bonus Percentage vs Salary

Higher bonus percentages tend to appear in higher-paid roles

The positive trend suggests that bonus structures scale with compensation. Because bonus percentage is itself part of the compensation architecture, this relationship is more direct than relationships involving demographic or skill variables.



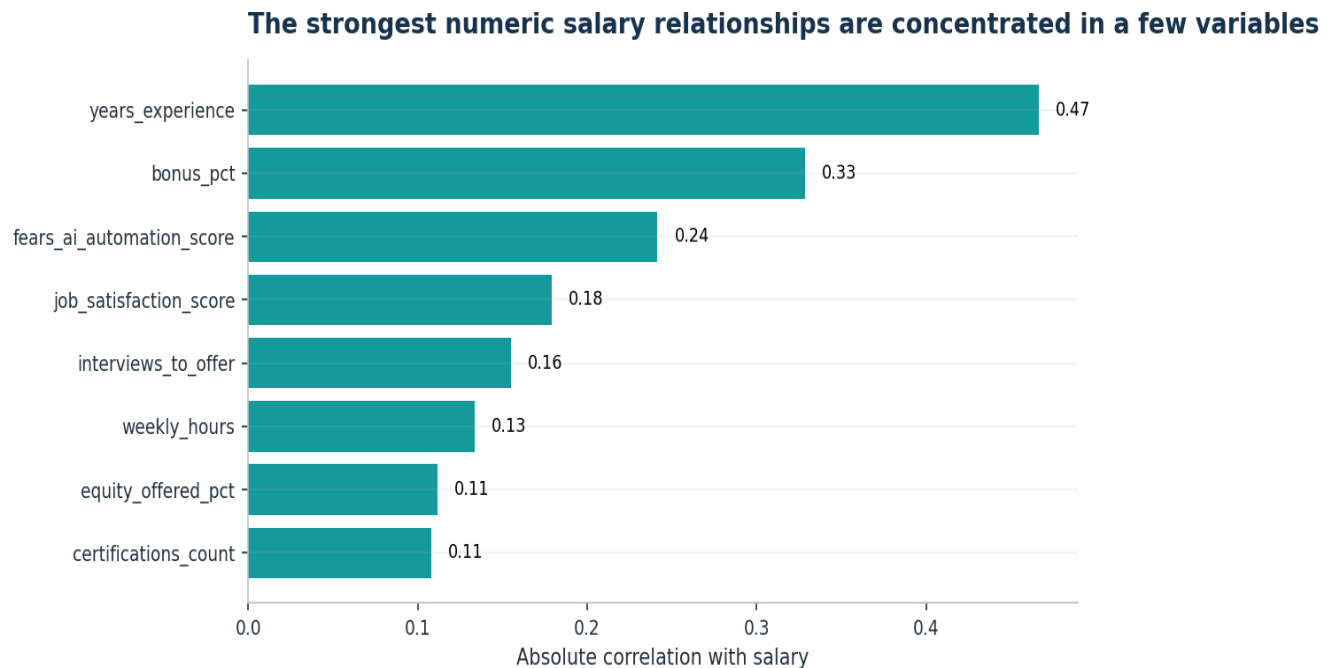
Key takeaways

- Bonus percentage correlation with salary: 0.33.
- High bonus percentages are concentrated among higher-paid observations, but there is still substantial spread.
- Base salary and total compensation should be distinguished when evaluating offers.

Final Findings & Practical Recommendations

What the dataset suggests for students, professionals and employers

The strongest numeric relationships are concentrated in experience, bonus structure, satisfaction and automation fear, while categorical comparisons show major differences by role, location, company size and management responsibility. For a student entering data science, the most defensible strategy is to build experience and role-relevant specialization rather than optimizing for one isolated tool.



Key takeaways

- Prioritize strong fundamentals plus practical experience: projects, internships and progressively complex work.
- AI/ML specialization is associated with higher pay in this sample, especially when paired with engineering or research responsibilities.
- Leadership and scope matter: people management is strongly associated with higher compensation.
- Use salary benchmarks by role and location rather than relying on one overall average.
- Treat AI adoption as a capability multiplier, not a salary guarantee.
- For employers, compensation analysis should control for seniority, location and role before comparing groups.